

# Device Lock Portal API

|          |                             |                |              |
|----------|-----------------------------|----------------|--------------|
| Filename | Device Lock Portal API.docx | Reference      |              |
| Version  | 1.0.0                       | Date           |              |
| Status   |                             | Classification | Confidential |

## History

| Status/Version/Date     | Author  | Comment        |
|-------------------------|---------|----------------|
| Stable/1.0.0/2026-07-15 | Lei.Gao | First version. |

# 1 Scope

This document is responsible for the Portal API interface design of WhaleOS lock management service.

# 2 Architecture

This Open API is consistent with the Lock Portal functionality

# 3 Specification

The Device Lock Portal API follows standard SaaS API specifications, including URL format, authentication mechanism, and response format.

## 3.1 URL

API URL follow the following specifications:

```
https://<host>/lock-portal/open/<api-version>/<api-path>
```

Host See the table below:

| Environment | Host                        |
|-------------|-----------------------------|
| ACC         | acc-lockservice.whaletv.com |
| PROD        | lockservice.whaletv.com     |

API Version is currently fixed as: v1

Example:

```
https://acc-lockservice.whaletv.com/lock-portal/open/v1/devices/import
```

## 3.2 Authentication

Each docking party needs to apply for a pair of credentials in the company's permission management system:

| field            | Comment                                                           |
|------------------|-------------------------------------------------------------------|
| <b>accessKey</b> | public key                                                        |
| <b>secretKey</b> | Secret key, properly stored, participate in signature calculation |

|         |          |
|---------|----------|
| brandId | Brand ID |
|---------|----------|

#### Request header to carry:

Authorization: <accessKey>:<signature>:<timestamp>

#### Among them:

accessKey: AccessKey obtained through application

timestamp: The current timestamp (in milliseconds) should not deviate from the server time by more than 1 hour

signature: Calculate according to the following algorithm

#### Signature algorithm:

signature = Base64(HmacSHA1(secretKey, timestamp + requestURI))

requestURI: Request path, without domain name or query string, for example:

`/log portal/open/v1/devices/import`

The server recalculates the signature according to the same rules and compares it. If there is any inconsistency, it will directly return 401

#### Signature Example(Java):

```
String accessKey = "your-access-key";
String secretKey = "your-secret-key";
String requestURI = "/lock-portal/open/v1/devices/import";
long timestamp = System.currentTimeMillis();

String data = timestamp + requestURI;
Mac mac = Mac.getInstance("HmacSHA1");
mac.init(new SecretKeySpec(secretKey.getBytes(StandardCharsets.UTF_8), "HmacSHA1"));
byte[] signBytes = mac.doFinal(data.getBytes(StandardCharsets.UTF_8));
String signature = Base64.getEncoder().encodeToString(signBytes);

String authorization = accessKey + ":" + signature + ":" + timestamp;
// Request header to join: Authorization: <authorization>
```

#### Authentication Failure Response:

| Message                                     | Comment                              |
|---------------------------------------------|--------------------------------------|
| <b>AUTHORIZATION EMPTY</b>                  | Not carrying Authorization header    |
| <b>AUTHORIZATION FORMAT ERROR</b>           | Format error                         |
| <b>AUTHORIZATION TS ERROR</b>               | Timestamp deviation exceeding 1 hour |
| <b>AUTHORIZATION ACCESSKEY NON-EXISTENT</b> | AccessKey does not exist             |
| <b>AUTHORIZATION SIGNATURE ERROR</b>        | Signature verification failed        |
| <b>USERINFO IS NULL</b>                     | No API permission                    |

### 3.3 Response

Successful response:

```
{
  "code": 0,
  "msg": "success",
  "data": {}
}
```

Error response:

```
{
  "code": error code,
  "msg": "error description"
}
```

## 4 API List

### 4.1 Get Device List

| API Path    | /open/v1/devices    |
|-------------|---------------------|
| Method      | GET                 |
| Description | Paging query device |

Parameters: Query String

| Name                | Required | Comment                                        |
|---------------------|----------|------------------------------------------------|
| brandId             | Y        | Brand ID                                       |
| mac                 | N        | MAC address                                    |
| sn                  | N        | serial number                                  |
| status              | N        | Lock status: 0-Unlocked, 1-Locked              |
| nextInstallmentDate | N        | Next installment payment date, yyyy-MM-dd      |
| dueStatus           | N        | Due status: 0- Not overdue, 1- Already overdue |
| paymentStatus       | N        | Payment status: 0-In Progress, 1-Complete      |
| pageNum             | N        | Page number, default value: 1                  |
| pageSize            | N        | Quantity per page, default value: 20           |

Response:

| Field | Type | Description |
|-------|------|-------------|
|-------|------|-------------|

|                            |         |                                                       |
|----------------------------|---------|-------------------------------------------------------|
| <b>id</b>                  | Long    | Device ID                                             |
| <b>mac</b>                 | String  | MAC address                                           |
| <b>sn</b>                  | String  | serial number                                         |
| <b>status</b>              | Integer | Lock status: 0-Unlocked, 1-Locked                     |
| <b>lastOnlineTime</b>      | Long    | Last online time of the device, millisecond timestamp |
| <b>importTime</b>          | Long    | Import time, millisecond timestamp                    |
| <b>nextInstallmentDate</b> | String  | Next installment payment date, yyyy-MM-dd             |
| <b>dueStatus</b>           | Integer | Due status: 0- Not overdue, 1- Already overdue        |
| <b>paymentStatus</b>       | Integer | Payment status: 0-In Progress, 1-Complete             |

## 4.2 Device Detail

|                    |                         |
|--------------------|-------------------------|
| <b>API Path</b>    | /open/v1/devices/detail |
| <b>Method</b>      | GET                     |
| <b>Description</b> | Get device details      |

Parameters: Query String

| Name           | Required | Comment   |
|----------------|----------|-----------|
| <b>id</b>      | Y        | Device ID |
| <b>brandId</b> | Y        | Brand ID  |

Response:

| Field                      | Type    | Description                                           |
|----------------------------|---------|-------------------------------------------------------|
| <b>id</b>                  | Long    | Device ID                                             |
| <b>brandId</b>             | Long    | Brand ID                                              |
| <b>mac</b>                 | String  | MAC address                                           |
| <b>sn</b>                  | String  | serial number                                         |
| <b>status</b>              | Integer | Lock status: 0-Unlocked, 1-Locked                     |
| <b>lastOnlineTime</b>      | Long    | Last online time of the device, millisecond timestamp |
| <b>importTime</b>          | Long    | Import time, millisecond timestamp                    |
| <b>nextInstallmentDate</b> | String  | Next installment payment date, yyyy-MM-dd             |
| <b>dueStatus</b>           | Integer | Due status: 0- Not overdue, 1- Already overdue        |
| <b>paymentStatus</b>       | Integer | Payment status: 0-In Progress, 1-Complete             |
| <b>installmentCycle</b>    | Integer | Installment period (unit: month)                      |
| <b>curPincode</b>          | String  | The currently generated PIN code                      |
| <b>curPasscode</b>         | String  | The currently generated Pass Code                     |
| <b>clearStatus</b>         | Integer | Clear status: 0-Default, 1-Cleaning                   |

## 4.3 PIN code history

|             |                                  |
|-------------|----------------------------------|
| API Path    | /open/v1/devices/pincode-history |
| Method      | GET                              |
| Description | Get PIN code usage history       |

Parameters: Query String

| Name    | Required | Comment   |
|---------|----------|-----------|
| id      | Y        | Device ID |
| brandId | Y        | Brand ID  |

Response:

| Field    | Type    | Description                                      |
|----------|---------|--------------------------------------------------|
| passcode | String  | Pass code                                        |
| pincode  | String  | PIN code                                         |
| date     | Long    | Usage date, millisecond timestamp                |
| status   | Integer | PIN code status: 0- Not used, 1- Issued, 2- Used |

## 4.4 Batch Import Devices

|              |                         |
|--------------|-------------------------|
| API Path     | /open/v1/devices/import |
| Method       | POST                    |
| Content-Type | application/json        |
| Description  | Batch import devices    |

Parameters:

| Name    | Required | Comment                                                                            |
|---------|----------|------------------------------------------------------------------------------------|
| brandId | Y        | Brand ID                                                                           |
| isSn    | Y        | 0-MAC, 1- serial number                                                            |
| items   | Y        | String[]<br>Equipment identification list, with a maximum of 1000 entries per time |

Example:

```
{
  "brandId": 100,
  "isSn": 0,
  "items": [ "AA:BB:CC:DD:EE:01", "AA:BB:CC:DD:EE:02" ]
}
```

Response:

| Field | Type | Description |
|-------|------|-------------|
|-------|------|-------------|

|                |        |                              |
|----------------|--------|------------------------------|
| <b>success</b> | String | Number of successful imports |
| <b>fail</b>    | String | Number of failed imports     |
| <b>batchId</b> | Long   | Import Batch ID              |

## 4.5 Update Device

|                     |                                    |
|---------------------|------------------------------------|
| <b>API Path</b>     | <b>/open/v1/devices/update</b>     |
| <b>Method</b>       | POST                               |
| <b>Content-Type</b> | application/json                   |
| <b>Description</b>  | Modify the information of a device |

Parameters:

| Name                       | Required | Comment                                       |
|----------------------------|----------|-----------------------------------------------|
| <b>brandId</b>             | Y        | Brand ID                                      |
| <b>id</b>                  | Y        | Device ID                                     |
| <b>status</b>              | Y        | Lock status: 0-Unlocked, 1-Locked             |
| <b>paymentStatus</b>       | N        | Payment status: 0-In Progress, 1-Complete     |
| <b>nextInstallmentDate</b> | N        | The next installment payment date, yyyy-MM-dd |
| <b>installmentCycle</b>    | N        | Installment cycle (months)                    |

Example:

```
{
  "brandId": 100,
  "id": 1234567890,
  "status": 1,
  "paymentStatus": 0,
  "nextInstallmentDate": "2025-12-10",
  "installmentCycle": 3
}
```

Response: N/A

## 4.6 Batch Lock/Unlock

|                     |                                    |
|---------------------|------------------------------------|
| <b>API Path</b>     | <b>/open/v1/devices/batch-lock</b> |
| <b>Method</b>       | POST                               |
| <b>Content-Type</b> | application/json                   |
| <b>Description</b>  | Batch locking/unlocking            |

Parameters:

| Name | Required | Comment |
|------|----------|---------|
|------|----------|---------|



|                |   |                                                                      |
|----------------|---|----------------------------------------------------------------------|
| <b>brandId</b> | Y | Brand ID                                                             |
| <b>macs</b>    | Y | String[]<br>MAC list, with a maximum of 1000 entries per transaction |
| <b>status</b>  | Y | Lock status: 0-Unlocked, 1-Locked                                    |

Example:

```
{
  "brandId": 100,
  "macs": ["AA:BB:CC:DD:EE:01", "AA:BB:CC:DD:EE:02"],
  "status": 1
}
```

Response: N/A

## 4.7 Generate pin code

|                     |                          |
|---------------------|--------------------------|
| <b>API Path</b>     | /open/v1/devices/pincode |
| <b>Method</b>       | POST                     |
| <b>Content-Type</b> | application/json         |
| <b>Description</b>  | Batch locking/unlocking  |

Parameters:

| Name            | Required | Comment                                |
|-----------------|----------|----------------------------------------|
| <b>brandId</b>  | Y        | Brand ID                               |
| <b>mac</b>      | Y        | Device MAC address                     |
| <b>passcode</b> | Y        | Pass Code, Generated by the device end |

Example:

```
{
  "brandId": 100,
  "mac": "AA:BB:CC:DD:EE:FF",
  "passcode": "ABCD1234"
}
```

Response:

| Field          | Type   | Description        |
|----------------|--------|--------------------|
| <b>pincode</b> | String | Generated PIN code |

## 4.8 Clean Device

|                 |                        |
|-----------------|------------------------|
| <b>API Path</b> | /open/v1/devices/clear |
|-----------------|------------------------|

|              |                         |
|--------------|-------------------------|
| Method       | POST                    |
| Content-Type | application/json        |
| Description  | Batch locking/unlocking |

Parameters:

| Name           | Required | Comment   |
|----------------|----------|-----------|
| <b>brandId</b> | Y        | Brand ID  |
| <b>id</b>      | Y        | Device ID |

Example:

|                                                     |
|-----------------------------------------------------|
| <pre>{   "brandId": 100,   "id": 1234567890 }</pre> |
|-----------------------------------------------------|

Response: N/A

## 5 Error Code List

Error code and description information

| ErrorCode     | Description                                              |
|---------------|----------------------------------------------------------|
| <b>270201</b> | Required parameters are missing or illegal               |
| <b>270202</b> | BrandId is not within the scope of the accessKey binding |
| <b>270203</b> | Batch operation quantity exceeds the limit               |
| <b>270102</b> | Device does not exist                                    |
| <b>270104</b> | MAC address format error                                 |
| <b>270103</b> | No available PIN code                                    |